

ADI_SoundBar_SDK_Host_Interface_ Protocol_Guide_EVAL-MELODY-8

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Preface

Thank you for purchasing the SoundBar SDK from Analog Devices, Inc. With a rich, powerful instruction set, floating point precision and high speed execution, it makes SHARC processors a popular choice for designers of signal processing systems.

The evaluation board is designed to be used in conjunction with the -Cross Core® Embedded Studio (CCES) development tools environment and reference platform with an ADSP-215xx SHARC processor. The CCES development environment aids advanced application code development and debug, such as:

- Create, compile, assemble, and link application programs written in C++, C, and assembly
- Load, run, step, halt, and set breakpoints in application programs
- Read and write data and program memory
- Read and write core and peripheral registers

Access to the processor from a personal computer (PC) is achieved through an external JTAG emulator.

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Product Overview

The product features SoundBar software development environment supporting:

- HDMI input interface for high definition audio Input
- Provision to develop and integrate Customer Post Processing in the SoundBar Audio signal chain

Purpose of This Manual

This “ADI_SoundBar_SDK_Host_Interface_Protocol_Guide_EVAL-MELODY-8” provides information about the SPI Host interface protocol for the ADSP-21569 processor.

The following protocols are described:

- For sending various commands and parameters from the Host to the DSP.
- For reading status information from DSP by the Host.
- For downloading the customer post processing codes from the Host to the DSP.
- For writing specific internal memory locations of the DSP by the Host.

- For reading specific internal memory locations of the DSP by the Host.

Intended Audience

The ADI_SoundBar_SDK_Host_Interface_Protocol_Guide_EVAL-MELODY-8.pdf is intended for use by a software team and customers, both externally and internally to Analog Devices, to develop and port codes on SHARC EVAL-MELODY-8.

The primary audience for this manual is a programmer who is familiar with Analog Devices processors.

Programmers who are unfamiliar with Analog Devices processors can use this manual, this manual assumes that the audience has a working knowledge of the appropriate processor architecture, instruction set, and C/C++ programming languages. But should supplement it with other texts (such as the ADSP-21569 SHARC Processor Hardware Reference and the ADSP-21569 SHARC Processor Programming Reference) that describe your target architecture.

Programmers who are unfamiliar with Cross Core Embedded Studio should refer to the CCES online help

Manual Contents

The manual consists of:

- Chapter 1, "Introduction"
Describes basic information covered in this document.
- Chapter 2, "SPI Interface Protocols"
This chapter provides detailed information of hand shaking between DSP and Host
- Chapter 3, "Power on Sequence from Boot to Play"
This chapter provides information about the power on sequence from Boot to Play in serial boot mode

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Supported Processors

This evaluation system supports Analog Devices ADSP-21569 processors.

Product Information

Product information can be obtained from the Analog Devices Web site.

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The Analog Devices Web site, www.analog.com, provides information about a broad range of products—analog integrated circuits, amplifiers, converters, and digital signal processors.

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LinkedIn: Network with the LinkedIn group, Analog Devices SHARC: <http://www.linkedin.com>

Alphabetic List of Abbreviations and Acronyms

Abbreviation/ Acronym	Description
DAC	Digital to Analog Converter
DAI	Digital Audio Interface
DPI	Digital Peripheral Interface
DSP	Digital Signal Processor
FLG	Flag
GPIO	General Purpose Input Output
I2S	Inter Integrated Sound Interface
IRQ	Interrupt request
ISR	Interrupt Service Routine
JTAG	Joint Test Action Group
MCLK	Master Clock
MOSI	Master OUT Slave IN
MISO	Master IN Slave OUT
SPI	Serial Peripheral Interface

1. Introduction

The ADSP-21569 uses an SPI protocol for communicating with the Host micro.

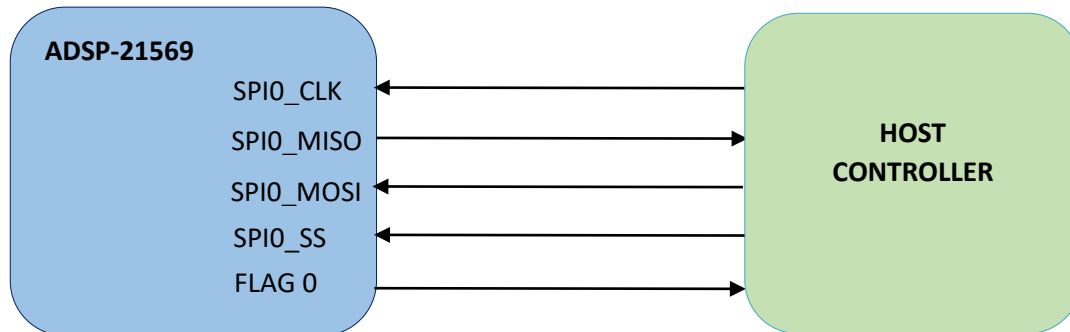


Figure 1: SPI Protocol for Communicating with Host Micro

- The ADSP-21569 communicates with the Host micro using an SPI bus
- The Host micro operates as the SPI bus master and ADSP-21569 operates as the SPI slave
- A memory buffer is defined in the ADSP-21569 processor where the command and parameters are stored
- When a command is executed, the ADSP-21569 stores the result of the execution in the command buffer and generates an interrupt
- The Host micro can then read the required location in the command buffer and determine if the command is successfully executed
- The status of play and decode operations is periodically placed in the command buffer by the ADSP-21569 processor
- To transfer a command and its associated parameters, the Host micro should transfer parameters followed by a command

1.1 Command Memory Map

Error! Reference source not found. shows the memory buffer structure for the command interface between the ADSP-21569 and the Host micro.

Table 1: Memory Buffer Structure for Command Interface

Command Buffer Location	Name	Variable Name in ADSP-21569
0x00	Command Register	Frmwk.Spi.COMMAND_BYTE
0x01	Parameter #1	Frmwk.Spi.PARAMETER_1
0x02	Parameter #2	Frmwk.Spi.PARAMETER_2
...	...	...
0x1f	Parameter #31	Frmwk.Spi.PARAMETER_31
0x20	Status #1	Frmwk.Spi.STATUS_BYTE1
0x21	Status #2	Frmwk.Spi.STATUS_BYTE2
0x22	Status #3	Frmwk.Spi.STATUS_BYTE3
0x23	Status #4	Frmwk.Spi.STATUS_BYTE4
...	...	
0x2f	Status #16	Frmwk.Spi.STATUS_BYTE16
0x30	Reserved	Frmwk.Spi.RESERVED_BYTE1
0x31	Reserved	Frmwk.Spi.RESERVED_BYTE2
...	...	...
0x3e	Reserved	Frmwk.Spi.RESERVED_BYTE15
0x3f	Interrupt Description	Frmwk.Spi.INTR_DESC_LOCN

- Apart from the above variables customer post processing can also access all the variables used in the ADI kernel by providing the addresses of those variables in the RAM space.
- Depending upon the requirements of customer post processing, the variable names and addresses of the required variables can be provided.

ADSP-21569 Interrupt Description

Read/Write Operation

- The ADSP-21569 provides an interrupt to the Host micro. Flag pin FLAG0 is used as an interrupt to the Host micro.
- The default level on this pin is logic LOW. Hence, the ADSP-21569 processor pulls this pin HIGH to interrupt the Host micro
- This pin is held HIGH until the Host micro performs a read operation
- When the Host micro receives an interrupt, it will perform a read on the interrupt description location
- A read operation should always be preceded by an address write operation
- As soon as the content of the addressed location is written to the SPI transmit buffer, the interrupt line is pulled LOW and the contents of the interrupt description location is cleared.
- Figure 2: Timing of Flag 0 with Respect to Interrupt Description Read (Not to scale) shows the timing of Flag0 going HIGH and LOW with respect to the interrupt description read

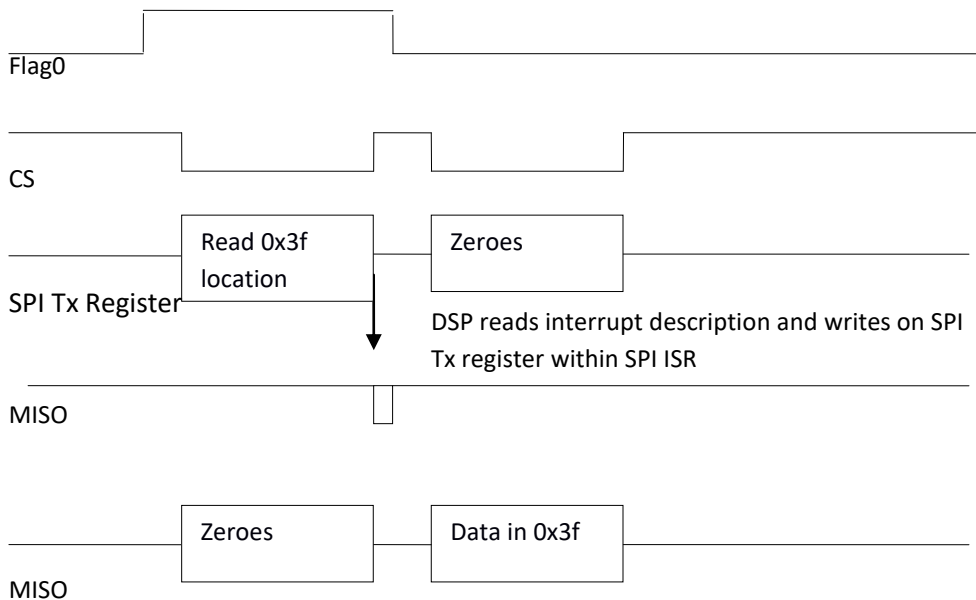


Figure 2: Timing of Flag 0 with Respect to Interrupt Description Read

Command Execution between DSP's and Micro

- The ADSP-21569 also provides status information by which the Host micro will know if the processor has completed the execution of the previous command or if the command execution is still in progress (BUSY).
- As soon as the command is received from the Host, the ADSP-21569 changes the interrupt description location to BUSY.

- After the command is executed, the result is then placed in the interrupt description location.
- So, after sending the command, the Host micro can read the interrupt description location. If this location contains a BUSY value, it indicates that the processor has received the command and the command processing is under progress.
- Once command execution is complete, the interrupt description location will be loaded with the command execution result.

Bit Allocation Handling Multiple Events

- There is a chance that more than one event occurs before the interrupt is serviced by the Host micro. So, a bit allocation concept is used to define the interrupt description content.
- Whenever an event occurs, the corresponding bits in the interrupt description register is SET. Refer to Table 2: Interrupt Description Bit Definition for the bit allocation for different events.

Example Sample Cases

As an example, consider the following two sample cases:

Sample case 1:

The DSP first interrupts the Host for a stream change immediately followed by an interrupt to the Host for a status change. If the Host services the interrupt only after the second interrupt has occurred, the DSP informs the Host that both a stream change and a status change have occurred (both bits 5 and 4 in the interrupt description will be set).

Sample case 2:

The DSP first interrupts the Host for a command execution successful immediately followed by an interrupt to the Host for a status change. If the Host services the interrupt only after the second interrupt has occurred, the DSP informs the Host that both a command execution successful as well as a status change have occurred (both bits 2 and 4 will be set).

In an example where a status change happens within a particular stream, the following sequence of operations occurs.

1. The DSP detects this status change.
2. Writes status change within the same stream (0xxx xxxx xx01 xxxx b, that is, make corresponding bits 1) on to the interrupt description location. The DSP makes FLAG0 HIGH to interrupt the Host micro.
3. The Host micro sends an interrupt description read.
4. The DSP reads the content of the interrupt description location and writes it onto the SPI TX register.
5. The DSP pulls FLAG0 LOW.

6. The Host micro sends dummy zeroes to receive the data in the SPI Tx register.

Table 2: Interrupt Description Bit Definition

Bits	Interrupt Description Contents	Meaning
Bit 0	0000 0000 0000 0000 b	Default content in interrupt description
	0000 0000 0000 0001 b	Initialization done
Bits 2-1	0xxx xxxxxxxx x00x b	Reserved for ADI
	0xxx xxxxxxxx x01x b	BUSY: Command execution in progress
	0xxx xxxxxxxx x10x b	Command execution successful
	0xxx xxxxxxxx x11x b	Command execution error
Bits 3	0xxx xxxxxxxx 0xxx b	Patch code download not allowed
	0xxx xxxxxxxx 1xxx b	Patch code download allowed
Bits 5-4	0xxx xxxx xx00 xxxx b	Reserved for ADI
	0xxx xxxx xx01 xxxx b	Status change within same stream
	0xxx xxxx xx10 xxxx b	Reserved for ADI
	0xxx xxxx xx11 xxxx b	Stream change and status change
Bits 9-6	Reserved for ADI	Reserved for ADI
Bits 14-10	Allotted to customer post processing	Allotted to customer post processing
Bit 15	Reserved for ADI	Reserved for ADI

1.2 Flow Chart for Informing Host Micro about Status Change

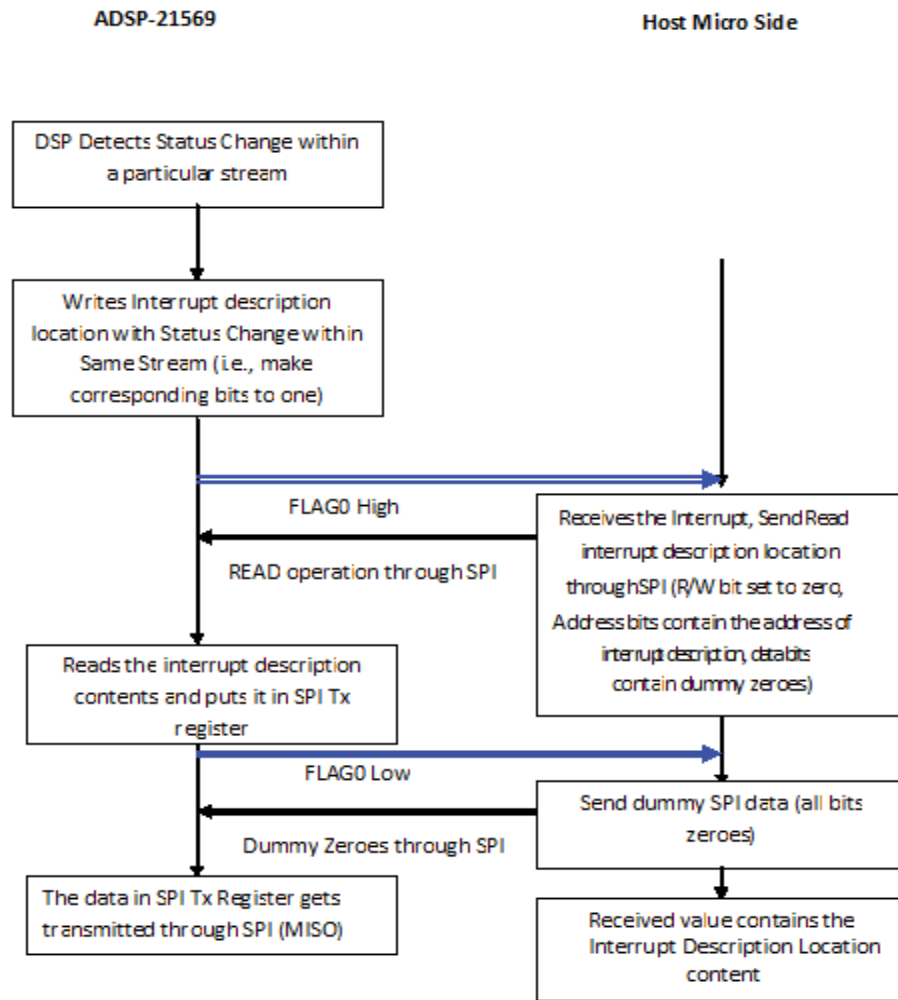


Figure 3: Flow Chart for Informing Host Micro about Status Change

The ADSP-21569 interrupts the Host micro under the conditions listed in Table 3: Interrupt Occurrence Sequence Description

Table 3: Interrupt Occurrence Sequence Description

No	Interrupt Occurrence Sequence Description
1	a. After power on, ADSP-21569 does all the hardware configurations, buffers and variables initialization and finally configures SPI registers. b. Writes Initialization Done to Interrupt Description Location. c. Sends interrupt to micro.
2	a. Micro sends Stop Play command to CORE. b. ADSP-21569 receives Stop Play command. c. ADSP-21569 processes Stop Play command, initializes corresponding internal memory variables. d. Writes Command Execution Successful to Interrupt Description Location. e. Sends interrupt to micro.
3	a. Micro sends Parameter #1 to #4 followed by General Decoding Options command to CORE1 b. ADSP-21569 receives Parameter #1 to #4 followed by command. c. After receiving General Decoding Options command, ADSP-21569 processes the command, initializes corresponding internal memory variables. d. Writes Command Execution Successful Error to Interrupt Description Location. e. Sends interrupt to micro.
4	a. Micro sends Sport Initialize command to CORE b. ADSP-21569 receives Sport Initialize command. c. ADSP-21569 processes Sport Initialize command, configures Sport if required and initializes corresponding internal memory variables. d. Writes Command Execution Successful to Interrupt Description Location. e. Sends interrupt to micro.
5	a. Micro sends Start Play command to CORE b. ADSP-21569 receives Start Play command. c. ADSP-21569 processes Start Play command, initializes corresponding internal memory variables. d. Writes Command Execution Successful to Interrupt Description Location. e. Sends interrupt to micro.
6	a. ADSP-21569 checks for any change in Status Information at the end of every frame. b. If there is any change, ADSP-21569 writes Status Change to Interrupt Description Location. c. Sends interrupt to micro.

2. SPI Interface Protocols

This chapter provides information about SPI interface between the Host and the DSP.



ALL SPI communication between the Host and the DSP are LSB first.

The following topics are covered:

- “Initial Handshake Between DSP and Host”
- “Sending Commands and Parameters from Host to DSP”
- “Reading Status Information from DSP by Host”

2.1 Initial Handshake between DSP and Host

2.1.1 Protocol

This section describes the initial handshake protocol between the DSP and the Host that starts further communication.

From Host Micro Side

1. Performs its own power on sequence.
2. When an interrupt from DSP comes, performs a read operation to the interrupt description location.
3. The Interrupt description location must contain Initialization done. If not, it resets the DSP.
4. If the Interrupt description location contains Initialization done, the Host can issue other commands.

From ADSP-21569 Side

1. Performs required software initializations, Program Flag pins, configures DSP as SPI slave.
2. Writes Initialization done to the interrupt description location.
3. Sends an interrupt to the Host micro by pulling FLAG0 HIGH.
4. When the Host sends the Read Interrupt Description, reads the data from the interrupt description location and writes it to the SPI Tx register. Pull FLAG0 LOW.
5. Receives and processes further commands.

2.1.2 Flow Chart

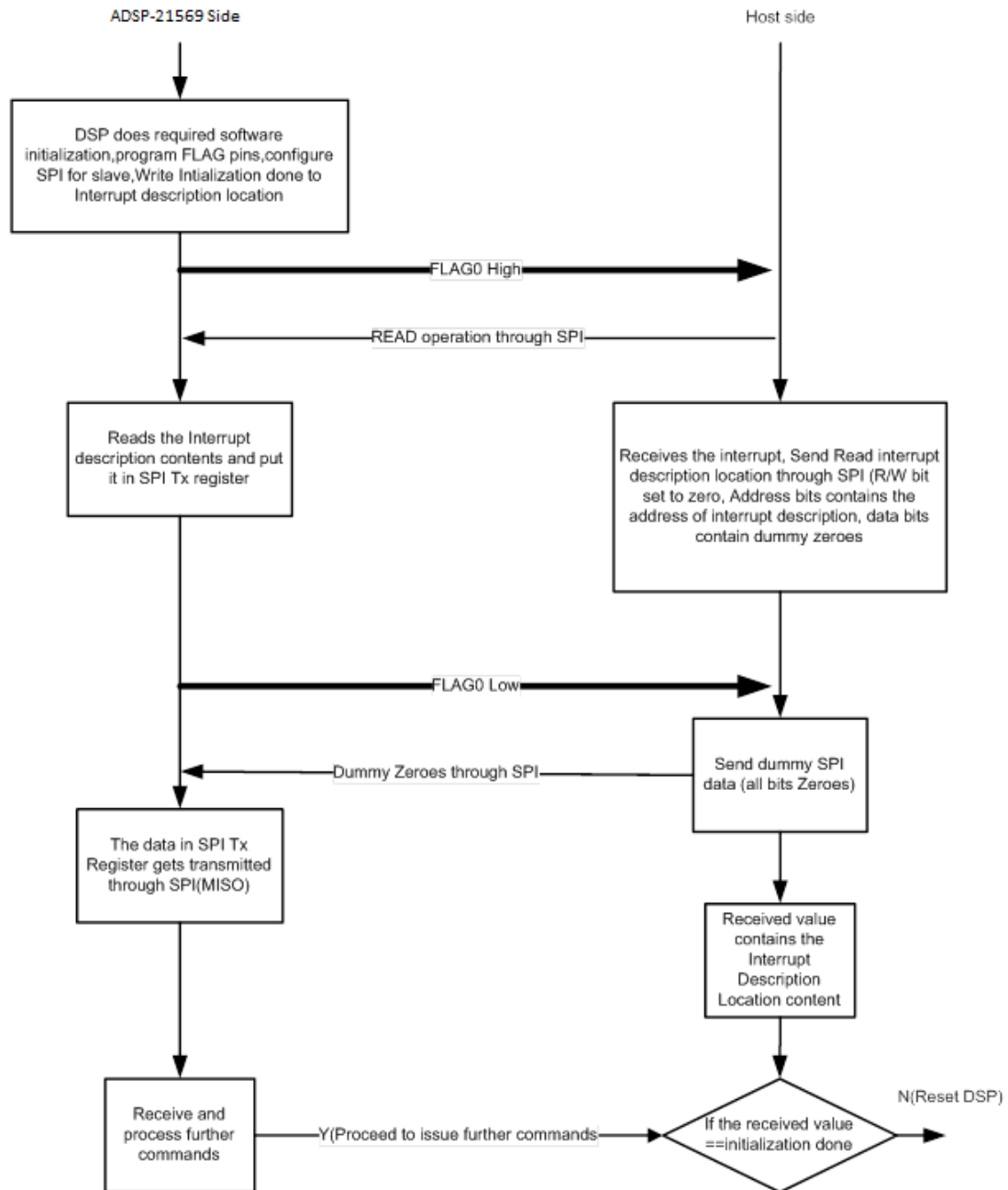


Figure 4: Initial Handshake between DSP and Host

2.1.3 SPI Communication

1. Host sends Read Interrupt description to ADSP-21569

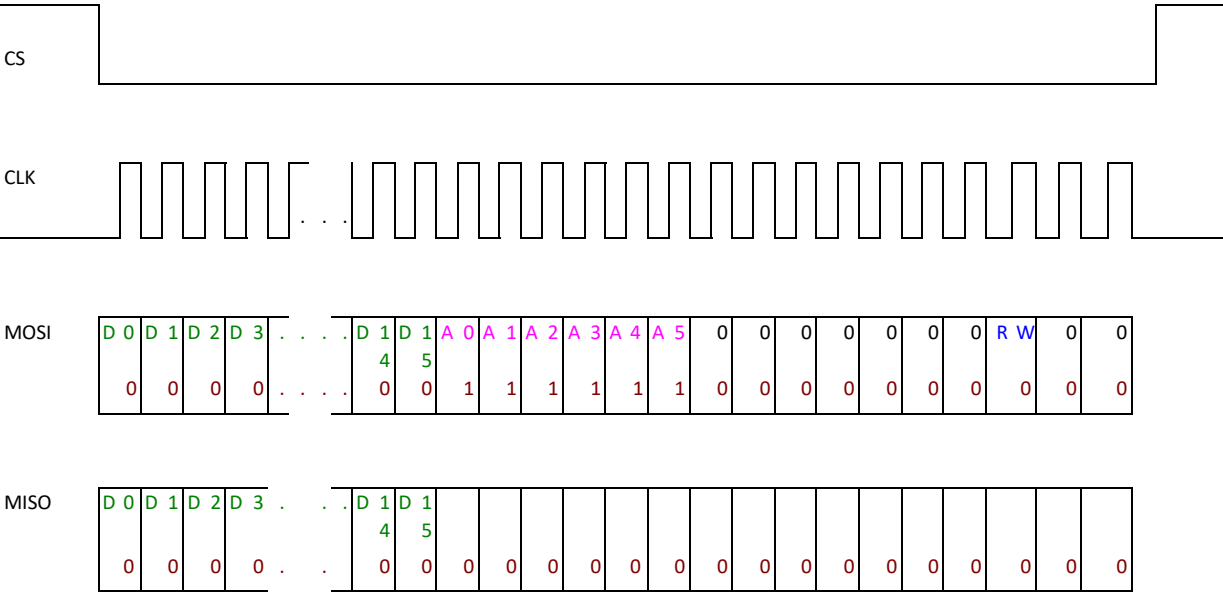


Figure 5: MOSI/MISO Information When Host Sends Read Interrupt Description Location

MOSI:

- R/W bit (bit 29) should contain 0 to indicate a read operation.
- A5 to A0 bits (bits 21 to 16) contain the address of the Interrupt Description (0x3f)
- D15 to D0 bits (bits 15 to 0) contain zero data.

MISO:

All bits have zeroes.

2. Host sends dummy zeroes to receive data from ADSP-21569

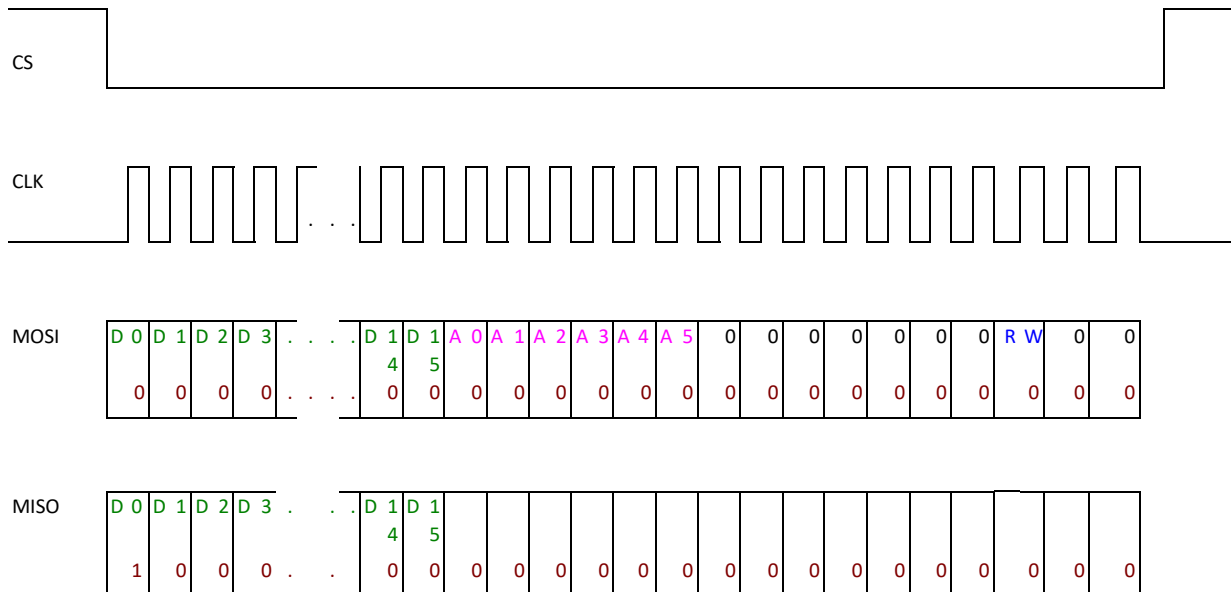


Figure 6: MOSI/MISO Information When Host Sends Dummy Zeros to Receive Data from DSP

MOSI:

All bits have dummy zeroes.

MISO:

D15 to D0 bits (bits 15 to 0) contain the actual data (for example, 0x0000 0001, indicating initialization done) present in the interrupt description location.

2.2 Sending Commands and Parameters from Host to DSP

2.2.1 Protocol

This section describes the protocol used for sending various commands and parameters from the Host to the core1.

From Host Micro Side

1. Writes parameters associated with command.
2. Writes command byte.
3. Waits for interrupt from ADSP-21569. After receiving an interrupt from the ADSP-21569, reads the interrupt description location, or, keeps reading the interrupt description location.
4. The contents of the location explain the command execution. If it contains Busy, then command execution is in progress. If the execution is successful, the Host micro can proceed to issue more commands if necessary. If the execution result is an error, the Host micro can issue the same command again. Or, the Host micro can reset the ADSP-21569 and start again.

From ADSP-21569 Side

1. The ADSP-21569 makes the interrupt description location Busy as soon as it receives the command in the SPI ISR.
2. In the foreground, the ADSP-21569 keeps polling the command location. As soon as this is non-zero, ADSP-21569 starts command execution.
3. After executing the command, the ADSP-21569 writes the result to the interrupt description.
4. Clears the command byte.
5. Issues interrupt to the Host through the FLAG0 pin.

2.2.2 Flow Chart

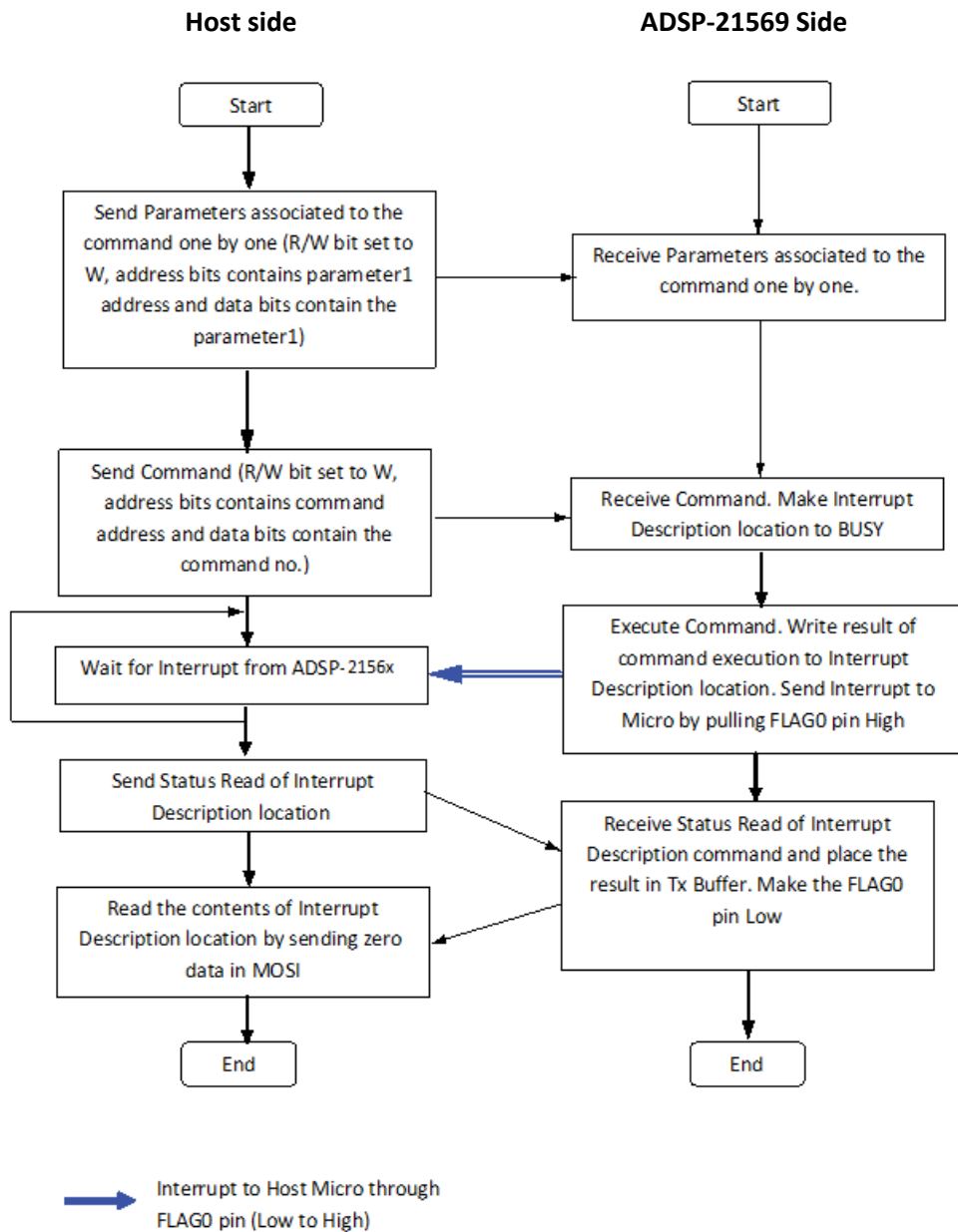


Figure 7: Sending Commands and Parameters from Host

2.2.3 SPI Communication

If the Host wants to send a command with a single parameter (command = 0x06, parameter1 = 0x05), the following SPI communication takes place for this operation.

Host sends Parameter1 to ADSP-21569.

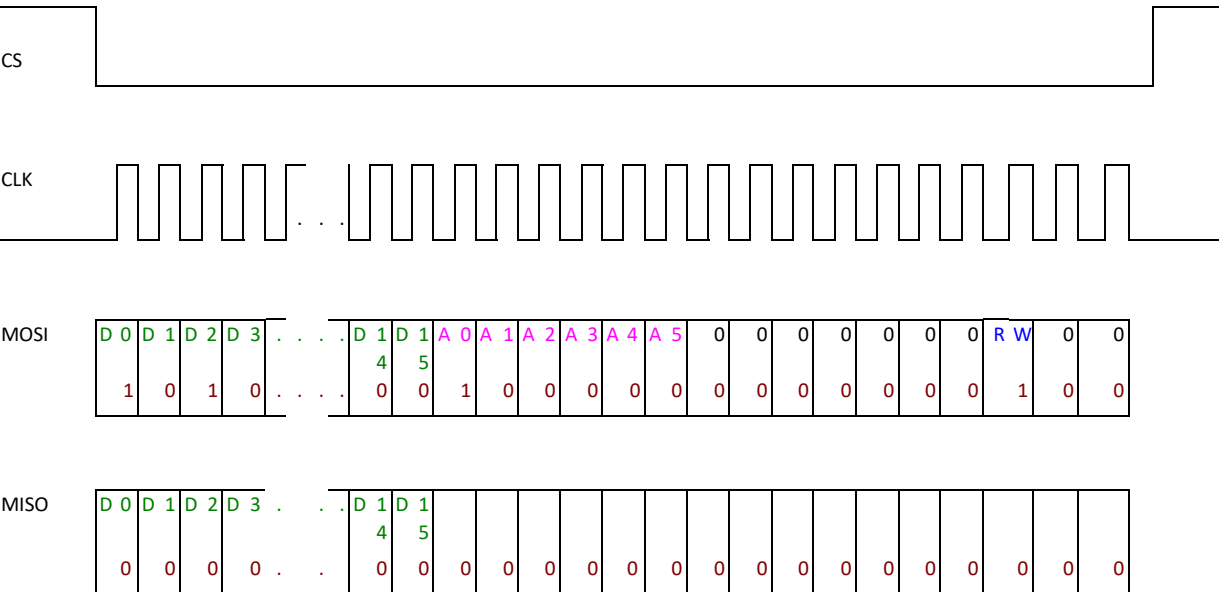


Figure 8: MOSI/MISO Information When Host Sends Parameter1 to DSP

MOSI:

- R/W bit (bit 29) should contain 1 to indicate a write operation.
- A5 to A0 bits (bits 21 to 16) contain the address of Parameter1 (0x01).
- D15 to D0 bits (bits 15 to 0) contain Parameter1 data (0x05).

MISO:

All bits have zeroes.

Timing diagram showing CS, CLK, MOSI, and MISO signals. The diagram is divided into two parts by an ellipsis.

CS (Chip Select): Active-low signal. High in the first part, low in the second part.

CLK (Clock): Periodic clock signal.

MOSI (Master Out Slave In): Data bus. In the first part, data 0110 is sent. In the second part, data 00000000000000001000 is sent.

MISO (Master In Slave Out): Data bus. In the first part, data 1010 is received. In the second part, data 00000000000000000000 is received.

MOSI:

A5 to A0 bits (bits 21 to 16) contain the address of the command (0x00).

MISO:

2.3 Reading Status Information from DSP by Host

This protocol can be initiated in the following ways:

- ### From Host Micro Side

- Page 25 of 32

4. This is repeated until it writes the address of the Last Status location.
5. Writes the address of the Last status location with the R/W bit set to R. The received value in MISO contains (Last –1) Status.
6. Sends zeroes.

From ADSP-21569 Side

1. ADSP-21569 reads the data present in the Status#1 location and writes to the Tx buffer.
2. The Status#1 information is transmitted and ADSP-21569 reads the data present in the Status#2 location and writes to the Tx buffer.
3. The Status#2 information is transmitted and ADSP-21569 reads the data present in the Status#3 location and writes to the Tx buffer.
4. Status information is transmitted until it reaches Last Status.
5. The (Last –1) Status information is transmitted and ADSP-21569 reads the data present in the Last Status location and writes to the Tx buffer.

The Last Status information is transmitted.

2.3.2 Flow Chart

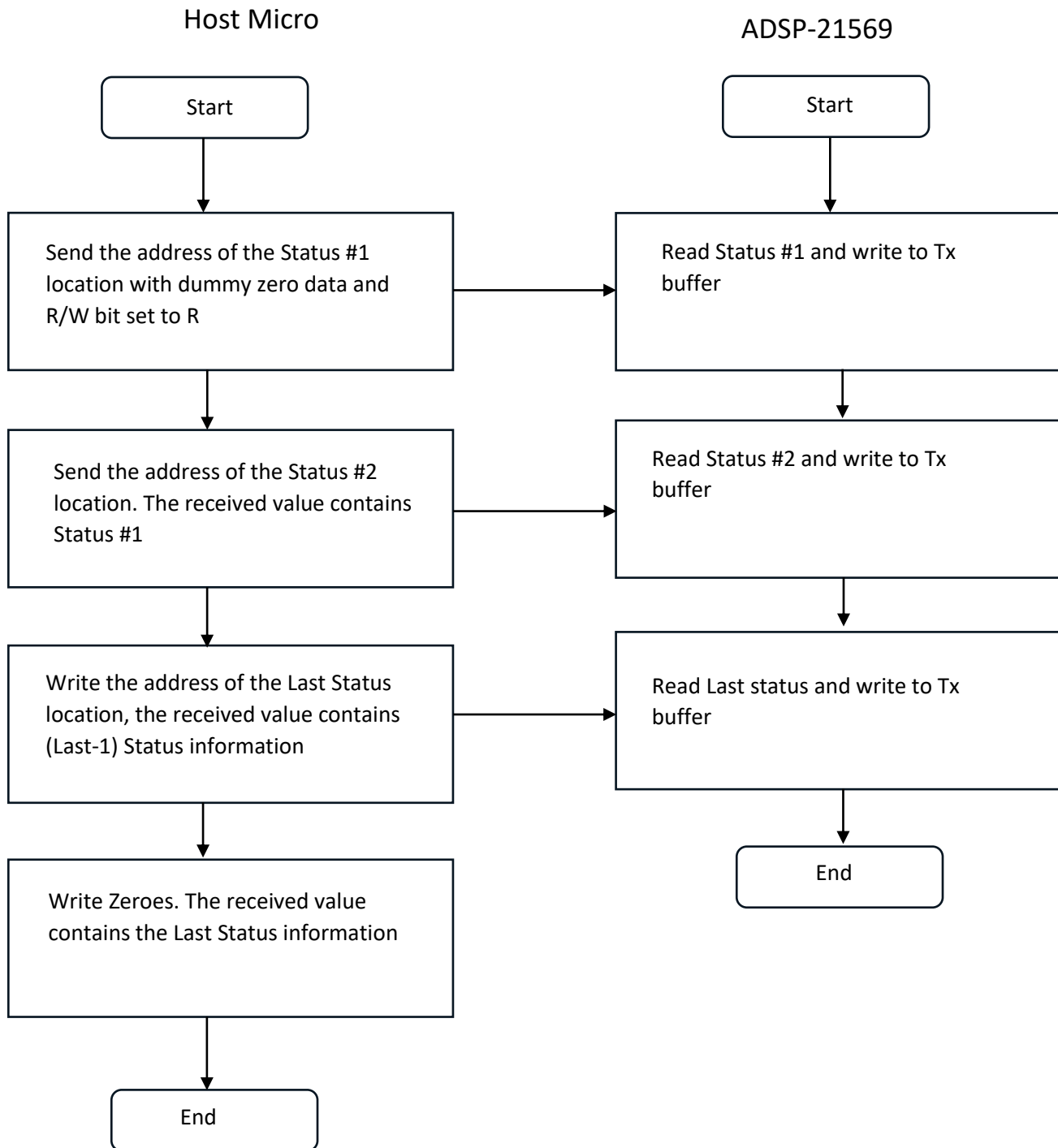


Figure 10: Reading Status Information from DSP by Host

2.3.3 SPI Communication

If the Host wants to read status words Status #1 and Status #2 from the DSP, the following SPI communication takes place for this operation where:

Status #1 = 0x01

Status #2 = 0x02

Host sends Read Status #1.

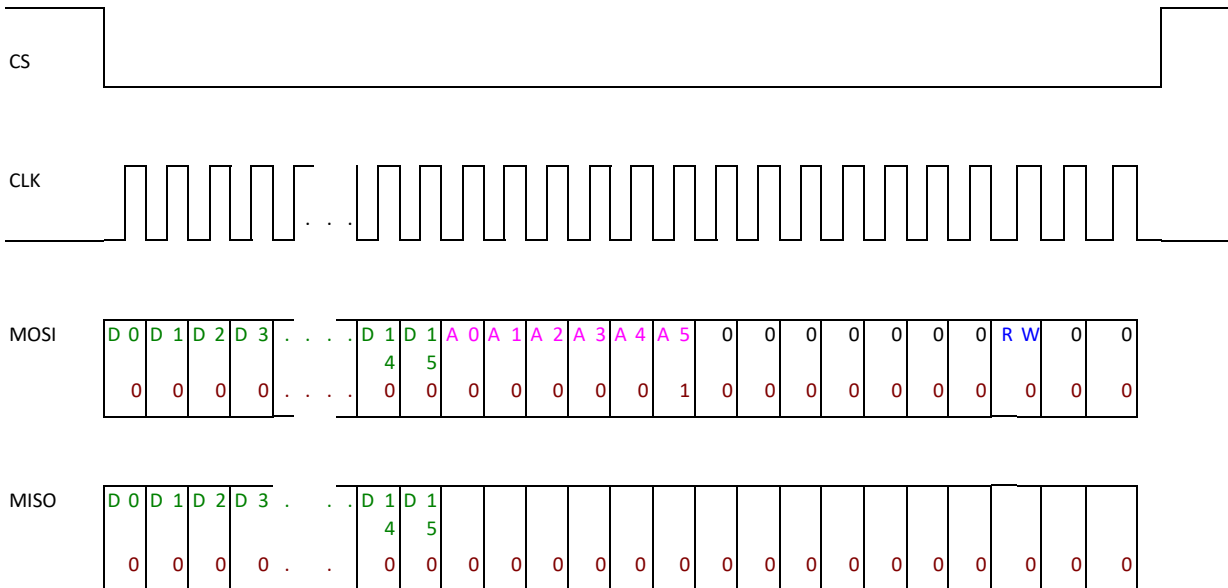


Figure 11: MOSI/MISO Information When Host Sends Read Status#1 Command

MOSI:

R/W bit (bit 29) should contain 0 to indicate a read operation.

A5 to A0 bits (bits 21 to 16) contain the address of the Status #1 (0x20).

D15 to D0 bits (bits 15 to 0) contain zero data.

MISO:

All bits have zeroes.

Host sends Read Status #2.

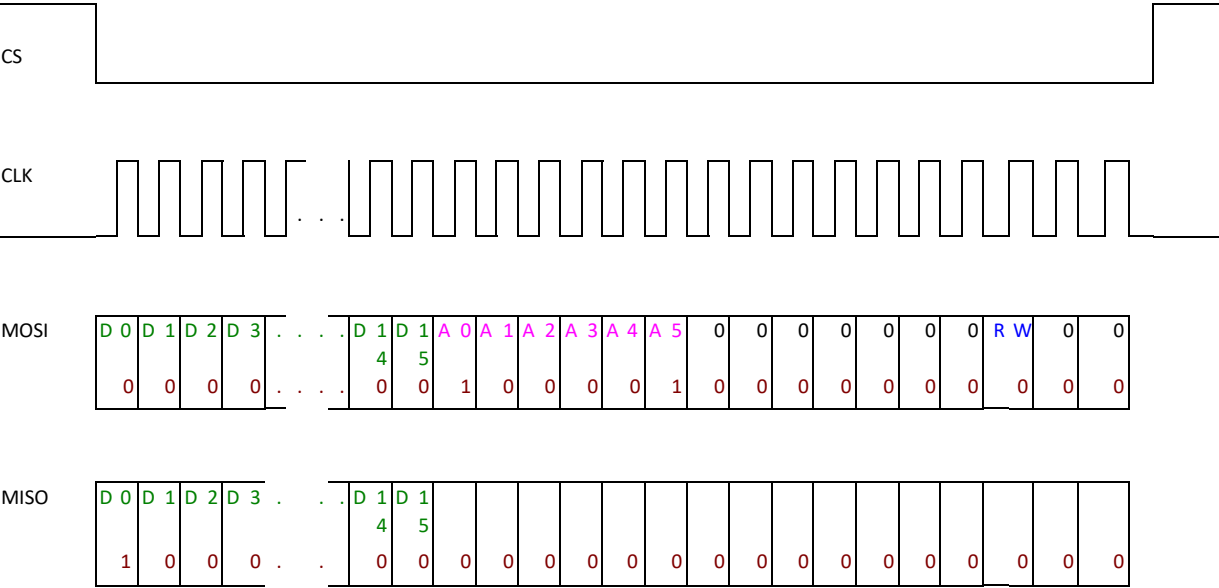


Figure 12: MOSI/MISO Information When Host Sends Read Status#2 Command

MOSI:

- R/W bit (bit 29) should contain 0 to indicate a read operation.
- A5 to A0 bits (bits 21 to 16) contain the address of the Status #2 (0x21)
- D15 to D0 bits (bits 15 to 0) contain zero data.

MISO:

- D15 to D0 bits (bits 15 to 0) contain Status#1 data (0x01)

Host sends dummy zeroes.

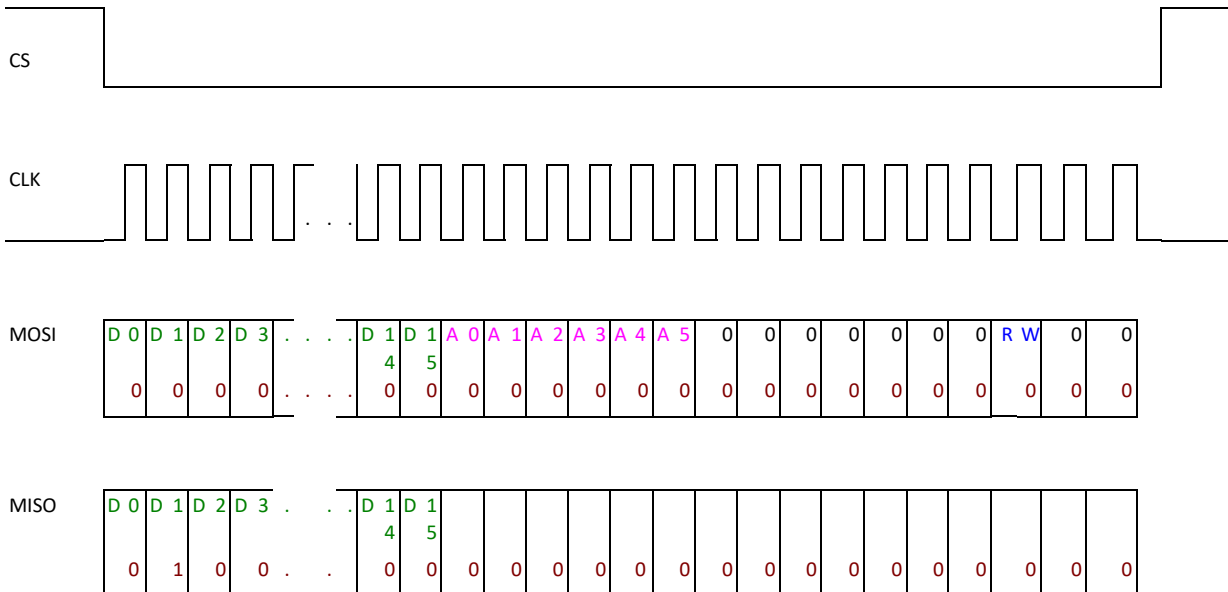


Figure 13: MOSI/MISO Information When Host Sends Dummy Zeros after Sending Status Read

MOSI:

All bits are zeroes.

MISO:

15 to D0 bits (bits 15 to 0) contain Status#2 data (0x02).

3. Power on Sequence from Boot to Play

This chapter provides information about the power on sequence from Boot to Play in serial boot mode.

The following topics are covered:

- “Flow Chart”
- “Sequence”

3.1 Flow Chart

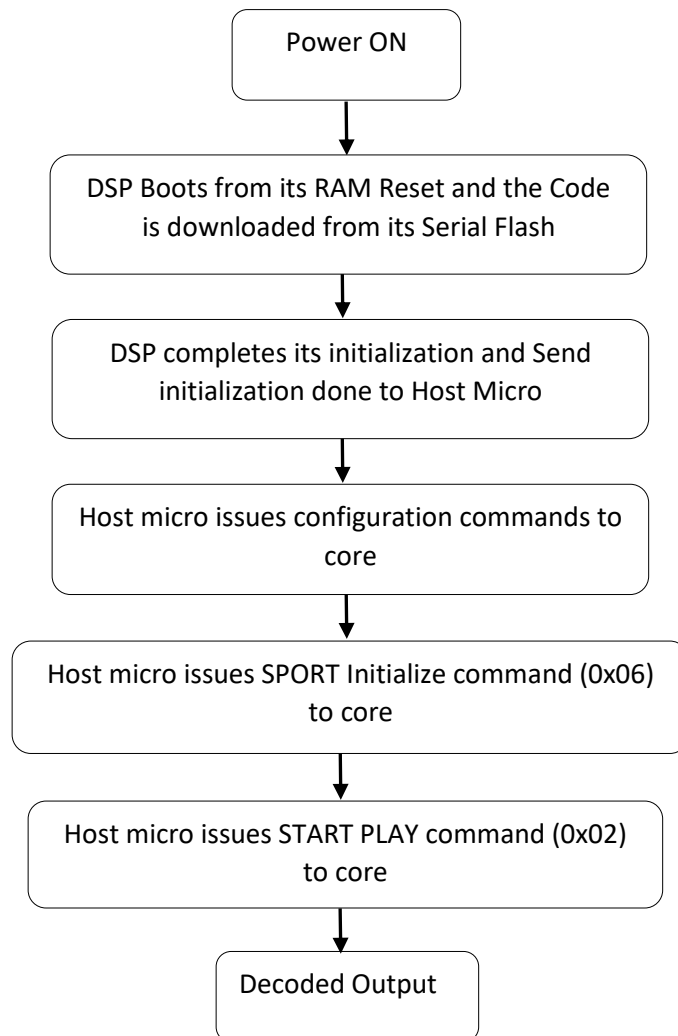


Figure 14: Power on Sequence from BOOT to PLAY

3.2 Sequence

If ADSP-21569 boots from its Flash connected to its Serial port, the following power on sequence takes place from Boot to Play.

Initialization:

1. Upon power on, ADSP-21569 boots from its internal RAM RESET and Download the Code from Serial Flash.
2. ADSP-21569 initializes heap memory, C Environment variables and registers.
3. ADSP-21569 initializes kernel variables and buffers.
4. ADSP-21569 configures the System Control register, Flag registers, SPI control registers, Sport control registers, SRU registers and Interrupt Control Registers.
5. ADSP-21569 enables global interrupts.
6. ADSP-21569 writes Initialization done to the interrupt description location and sends an interrupt to the Host micro.
7. The Host micro reads the interrupt description and understands that DSP has finished its initializations.

Configuration commands:

8. The Host micro can issue configuration commands for core. These are general decoding options, decoder and post decoder specific commands.
9. The DSP(s) will process each command and at the end of servicing the command, DSP(s) sends the result of the command execution to the Host micro.

Sport initializes command:

10. The Host micro can issue the Sport Initialize command (0x6) for core.
11. Depending on the auto detect mode set, corresponding Sport interrupts are enabled by the DSP1 and the result of the command execution is sent to the Host micro.
12. The incoming samples are received in the input DMA buffers and auto detection is performed on the incoming samples. Once a particular stream is detected, the stream type is informed to the Host micro.

Play command:

13. The Host micro can issue the Start Play command (0x02) for core in the sequential order. There is no need to wait for the DSP to inform the stream type.
14. Once the Start Play command is given, DSP(s) sends a command execution result to the Host micro and then decodes the input stream and outputs the decoded samples.